Language Endangerment Report

1. Dataset Overview

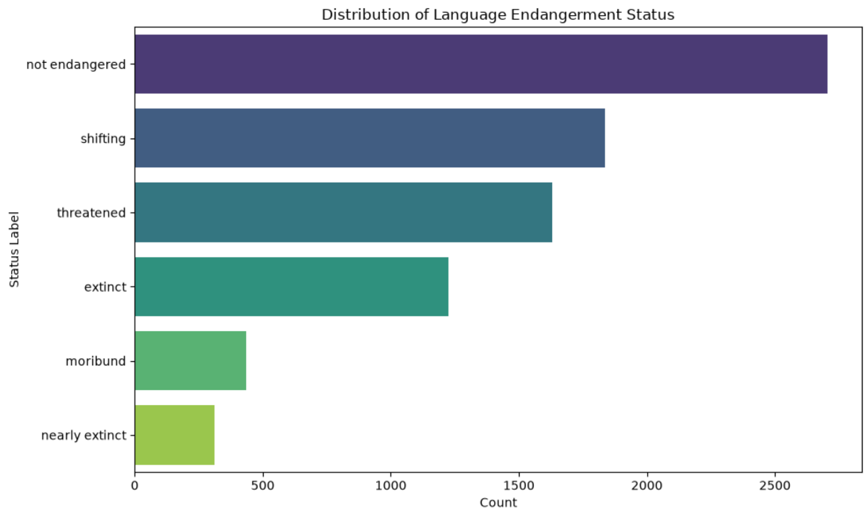
This dataset contains information on 8,612 languages worldwide, capturing 12 features such as geographic locations, language families, macroareas, and endangerment statuses. It serves as a comprehensive baseline for analyzing global linguistic diversity and vulnerability.

2. Data Quality Notes

- ISO-639-3 codes are missing for 8.8% of the dataset, which may complicate cross-referencing with other language databases.
- Endangerment status is missing for 5.5% of the recorded languages, representing a minor analytical blind spot.
- Latitude and longitude coordinates are unavailable for roughly 3.6% of the languages, affecting spatial mapping for these entries.
- Macroarea classifications are unassigned for 2.6% of the languages.
- Language family data is missing for 2.1% of the entries, which may impact family-level aggregation.

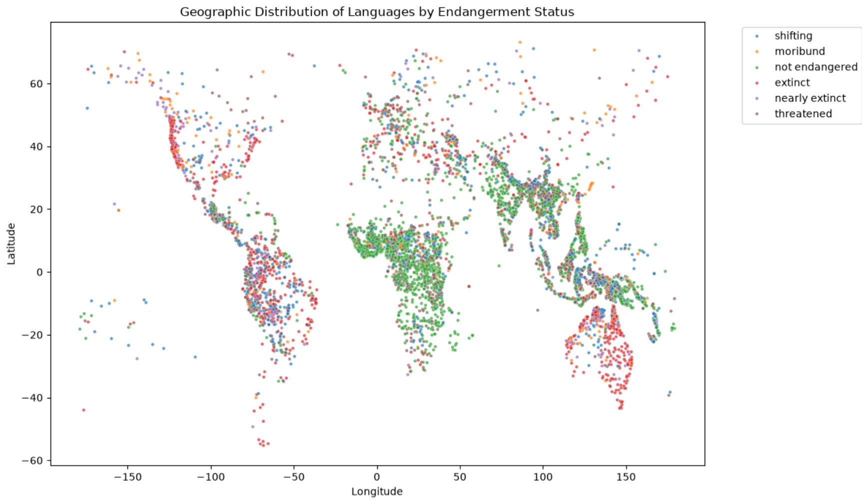
3. Exploratory Analysis

Chart 1: Language Endangerment Status Distribution



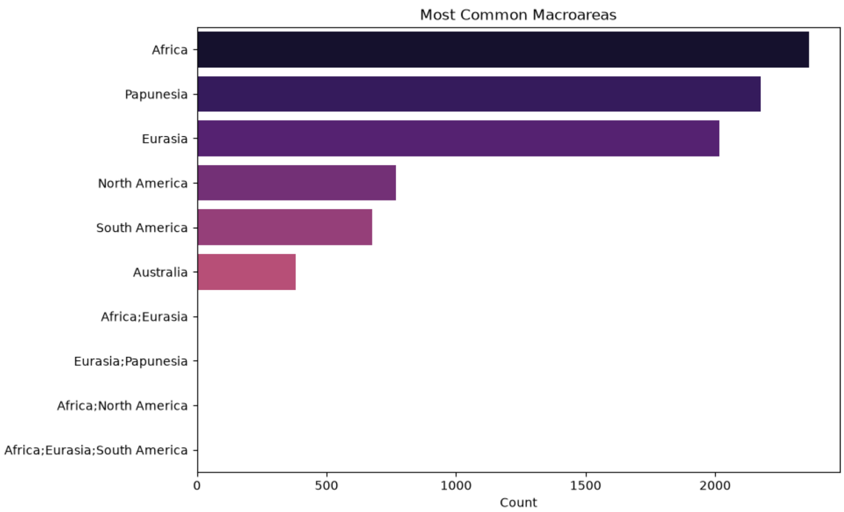
What it shows: Most languages in the dataset are classified as 'not endangered', but many face varying levels of endangerment.

Chart 2: Geographic Clusters of Languages



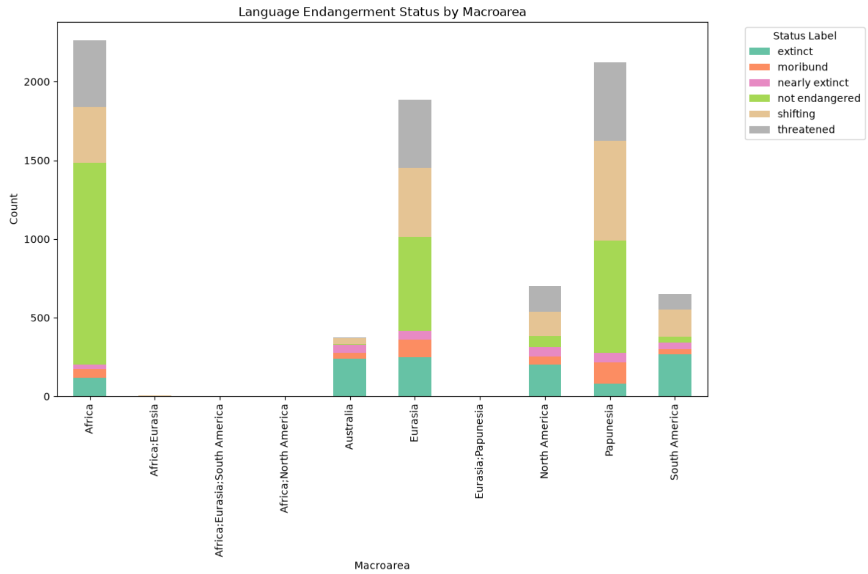
What it shows: Languages form distinct geographic clusters with noticeable spatial patterns for endangered versus healthy languages.

Chart 3: Macroarea Distribution



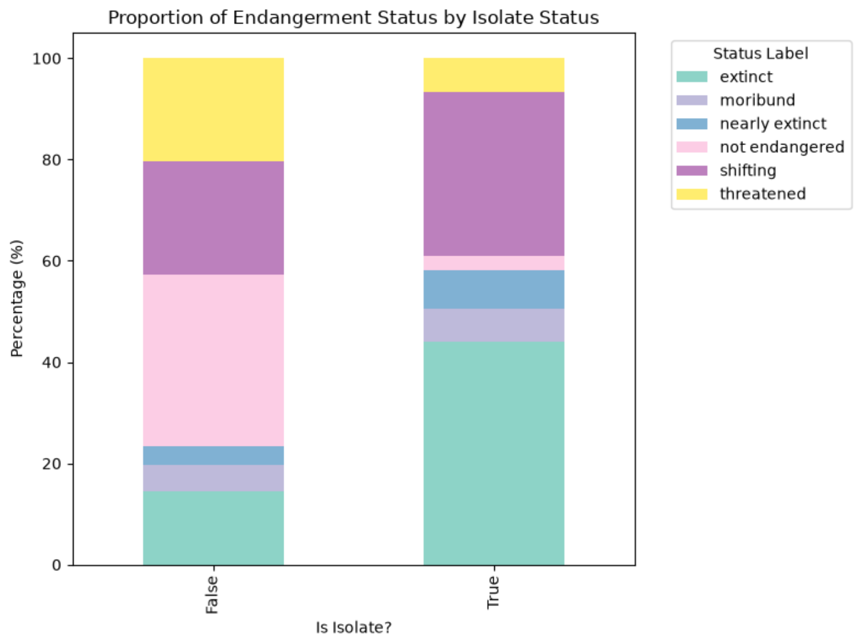
What it shows: Africa and Papunesia are the most commonly represented macroareas in this dataset.

Chart 4: Endangerment Status across Macroareas



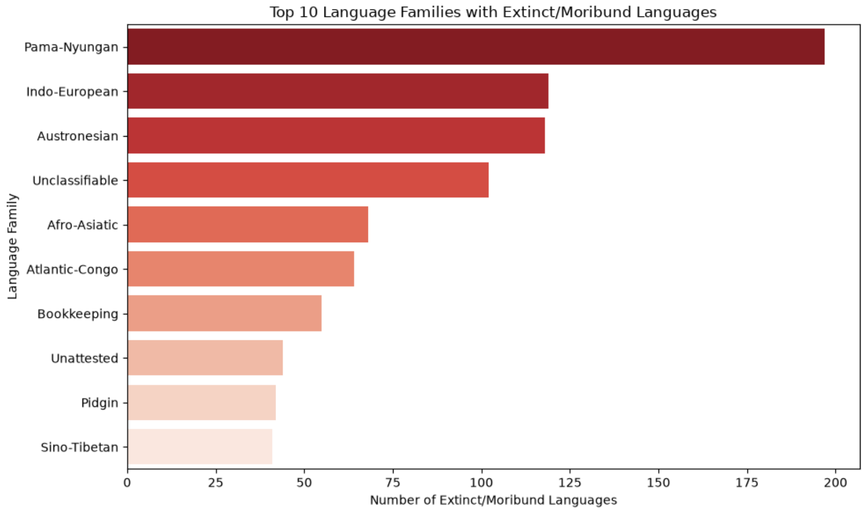
What it shows: The Americas and Australia display disproportionately high numbers of extinct or severely endangered languages.

Chart 5: Endangerment Status vs Isolate Status



What it shows: Language isolates tend to have a noticeably higher proportion of extinct and moribund statuses.

Chart 6: Language Families with Highest Endangerment



What it shows: Pama-Nyungan and Austronesian families have the highest absolute counts of highly endangered or extinct languages.

4. Modeling Results

A baseline Random Forest classifier was trained to predict the endangerment status. It achieved an overall accuracy of **0.4926**.

Class	Precision	Recall	F1-Score	Support
extinct	0.60	0.65	0.63	245
moribund	0.10	0.13	0.11	87
nearly extinct	0.14	0.21	0.17	62
not endangered	0.66	0.61	0.63	541
shifting	0.47	0.43	0.45	367
threatened	0.40	0.40	0.40	326
accuracy			**0.49**	**1628**
macro avg	0.40	0.40	0.40	1628
weighted avg	0.51	0.49	0.50	1628

5. Key Insights

- While "not endangered" is the most frequent individual status, the majority of the 8,612 languages collectively fall under some category of endangerment, with statuses like "shifting" and "threatened" forming a substantial at-risk middle ground.
- Distinct global language hotspots emerge in regions like Central Africa and Papua New Guinea, where healthy languages cluster tightly, whereas extinct or moribund languages show their own distinct regional densities.
- Africa dominates the dataset as the most linguistically diverse macroarea with 2,363 languages, followed closely by Papunesia, indicating highly localized global linguistic density.

- Australia and the Americas face acute linguistic crises, exhibiting disproportionately large shares of extinct or nearly extinct languages compared to other regions like Africa, which maintains a much healthier linguistic profile.
- Language isolates are significantly more vulnerable to extinction than languages belonging to larger families, with moribund and extinct statuses consuming a much larger percentage of the isolate category.
- The Pama-Nyungan and Austronesian families suffer from the highest absolute number of highly endangered or extinct languages, demonstrating that even widely distributed language groups are not immune to severe linguistic loss.
- A baseline Random Forest model struggles to accurately classify intermediate endangerment stages like "moribund" and "nearly extinct" (F1-scores of 0.11 and 0.17), limiting overall predictive accuracy to just 49.2%.

6. Recommendations / Next Steps

- Prioritize linguistic preservation and documentation efforts for language isolates and regions experiencing acute linguistic loss, specifically the Americas and Australia.
- Enhance predictive modeling by engineering advanced geographic or family-based features to better distinguish between intermediate endangerment categories (like moribund and nearly extinct).
- Investigate and impute missing geographic coordinates and status labels to construct a more accurate and comprehensive map of global linguistic vitality.